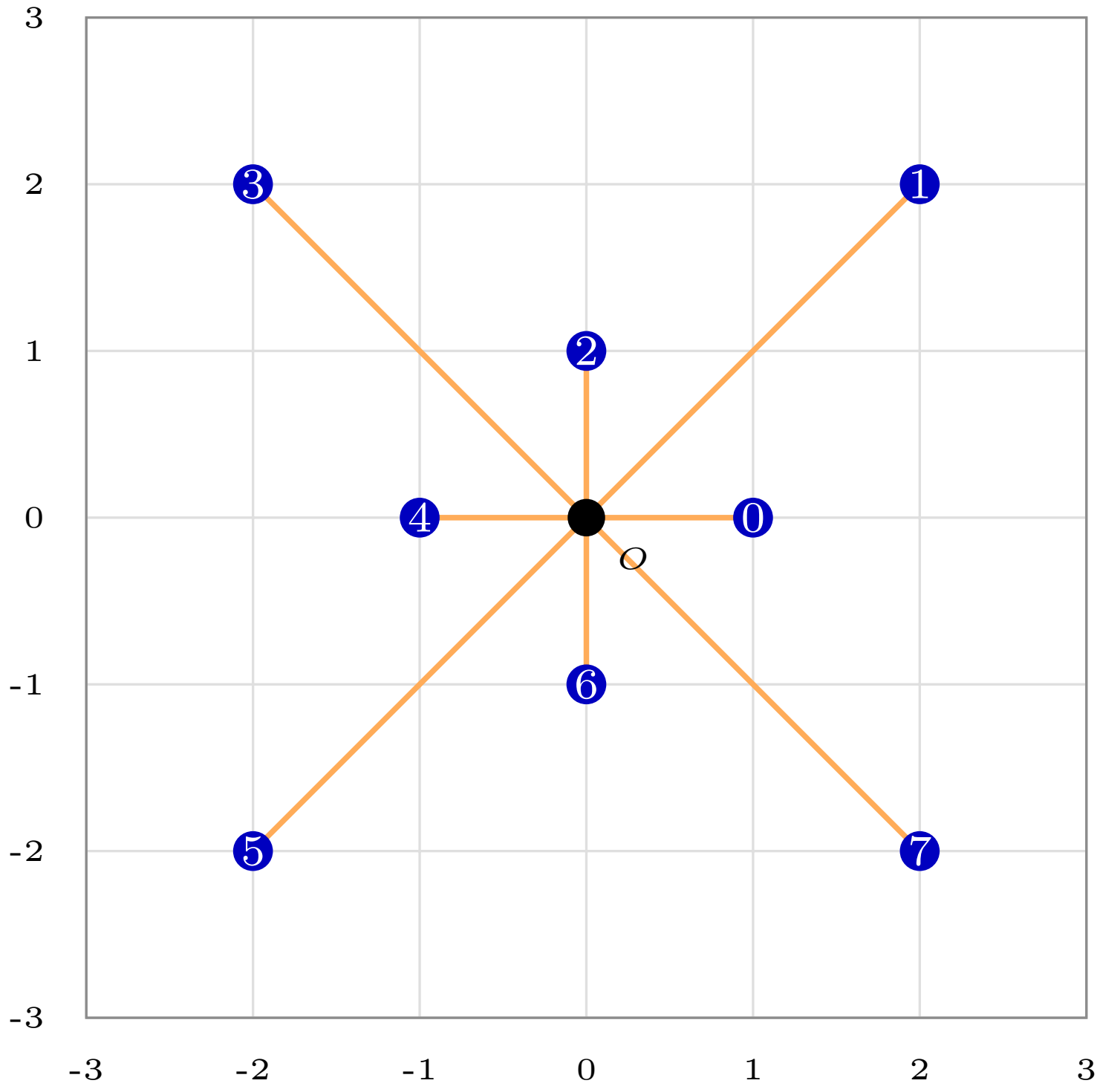
centre and antipodal pairs



the four chords meet at the centre $O = (0, 0)$;
k and $k + 4$ are antipodal